

Digital Image Processing

Textbook: Digital Image Processing, by Gonzalez and Woods



Digital Image Processing

- An image is a two-dimensional function $f(x,y)$, where x and y are the **spatial** (plane) coordinates, and the amplitude of f at any pair of coordinates (x,y) is called the intensity of the image at that level.
- If x,y and the **amplitude** values of f are **finite** and **discrete quantities**, we call the image a **digital image**.
- The field of digital image processing refers to processing digital images by means of a digital computer.
- A digital image is composed of a finite number of elements called **pixels**, each of which has a particular location and value.





**First Digital
Photograph
(1957)**



Levels of Image Processing

Low-Level Processes:

- Input: Image ->Output: Image
- Examples: Noise reduction, contrast enhancement, image sharpening.

Mid-Level Processes:

- Input: Image -> Output: Attributes (e.g., edges, contours, identity of objects)
- Examples: Segmentation (partitioning an image into regions/objects), description/classification of objects.

High-Level Processes:

- Input: Attributes -> Output: Understanding/Making sense
- Examples: "Making sense" of an ensemble of recognized objects (e.g., Scene understanding, Autonomous navigation).



The Origins of Digital Image Processing

- Newspaper industry (1920s): The Bartlane cable picture transmission system used to transmit images across the Atlantic.
- Reduced transmission time from more than a week to less than three hours.
- Images were coded using 5 levels of gray, later increased to 15 levels.
- The modern era began during the space race(1960), when images transmitted from space probes suffered from severe distortions. Scientists used computers to enhance and restore these images,
- Soon after, medical imaging transformed healthcare(1970). Techniques such as CT scanning allowed computers to reconstruct internal body structures from multiple projections.
- This highlighted a critical truth: **machines can reveal visual information that humans cannot directly perceive.**



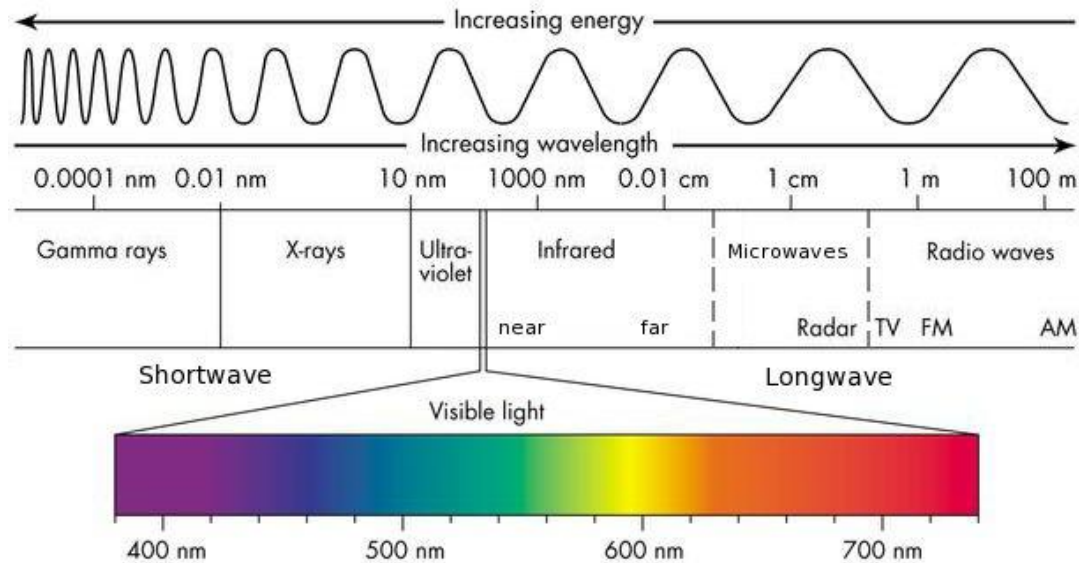
Examples of Fields that Use Digital Image Processing

- **Categorization by Source:**
- **Electromagnetic (EM) Spectrum:** Gamma-ray, X-ray, Ultraviolet, Visible, Infrared, Microwave, Radio waves.
- **Acoustic:** Ultrasound, Seismic.
- **Ultrasonic**
- **Electronic:** Electron Microscopy.
- **Computer-Generated:** Synthetic images (Fractals, 3D models).

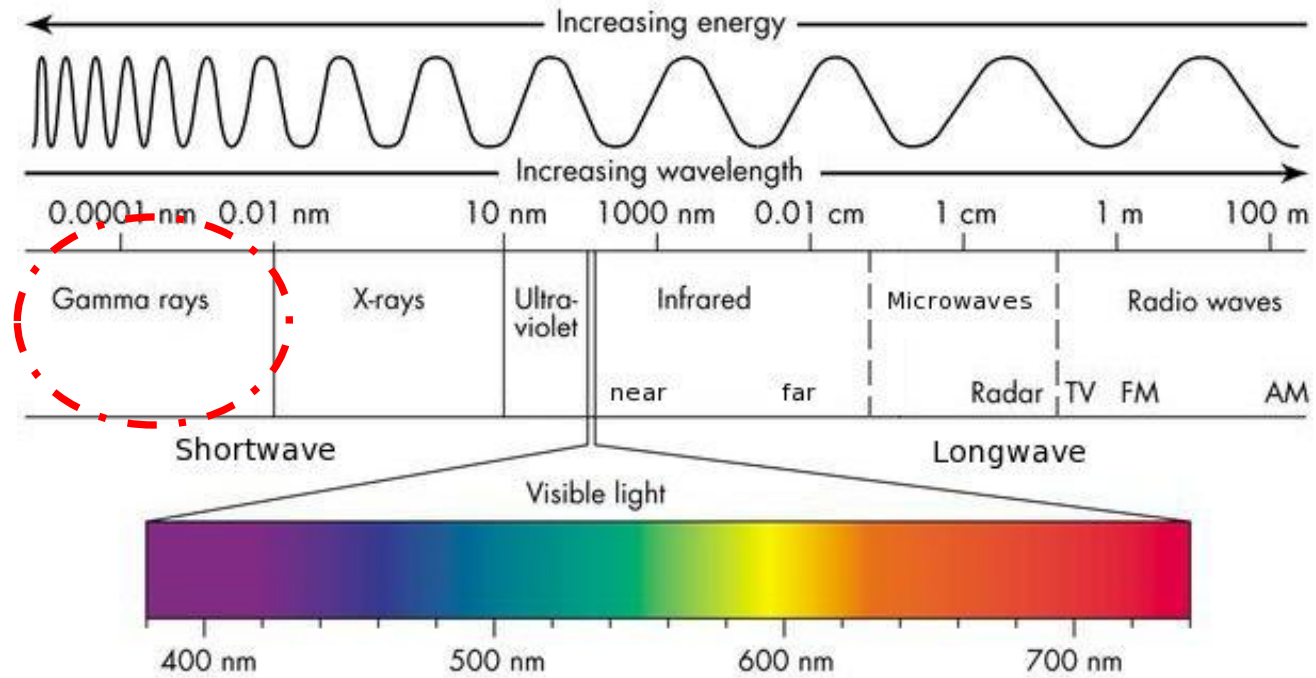


Sources of Digital Images

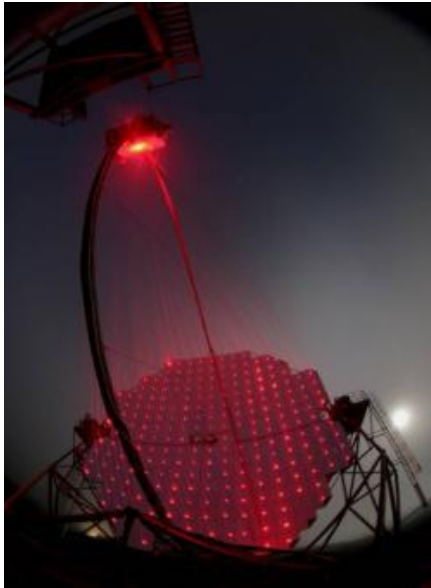
*The principal source for the images is the **electromagnetic (EM) energy spectrum**.*



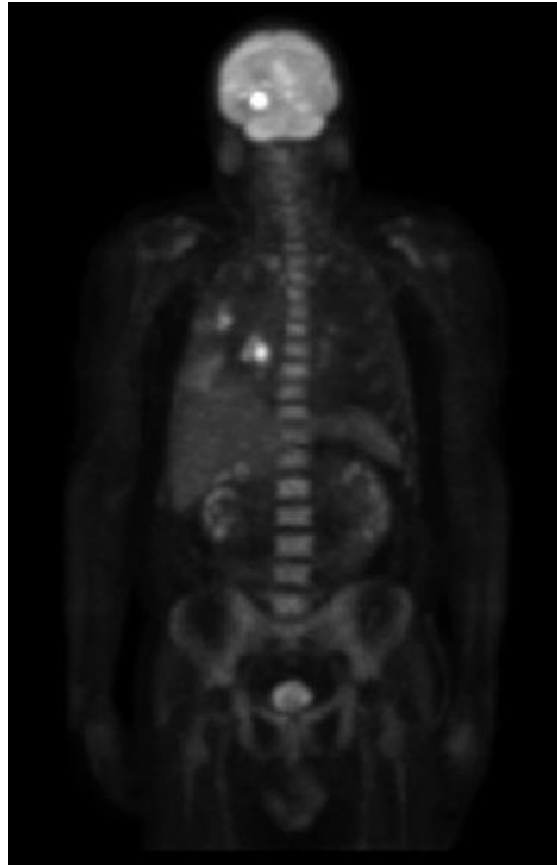
Gamma rays



Gamma rays



**Gamma-Ray
Imaging
Cherenkov
Telescope**



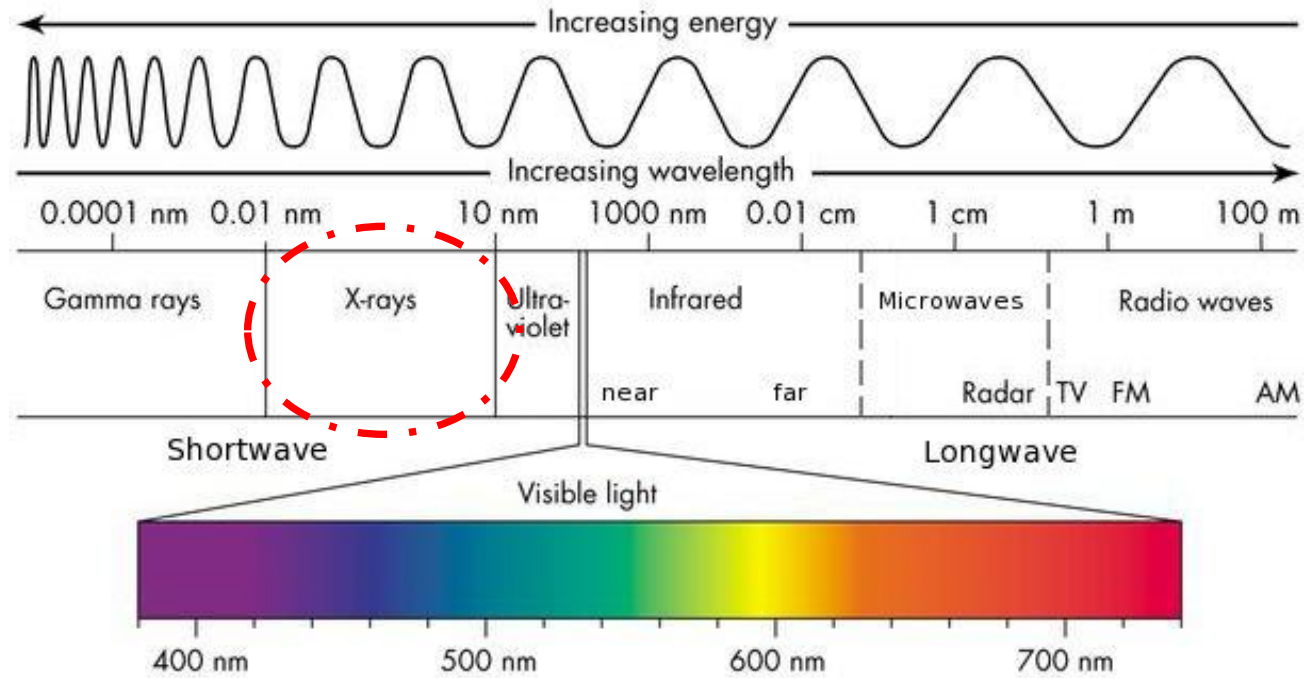
**Gamma-Ray Imaging
In nuclear medicine**



**Gamma-Ray imaging of
A starburst galaxy about
12 million light-years
away**



X- rays



X- rays

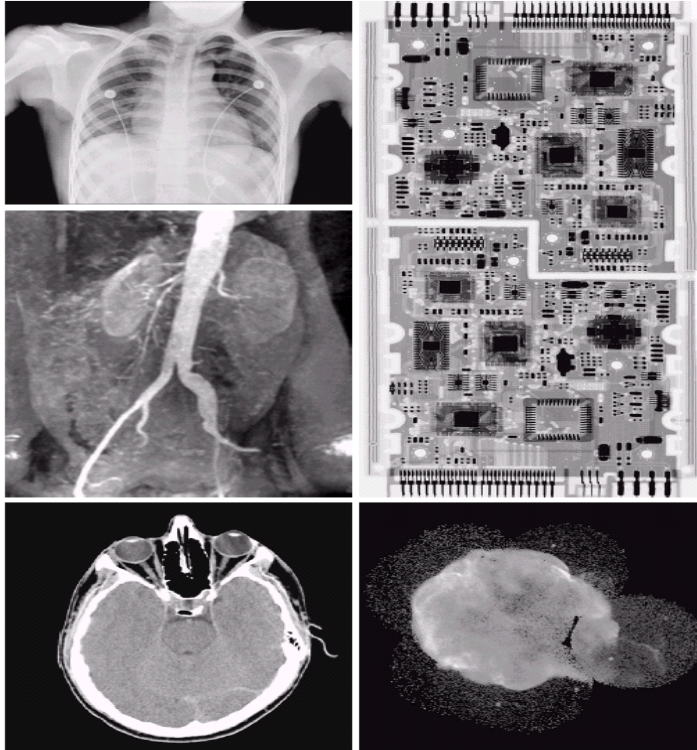
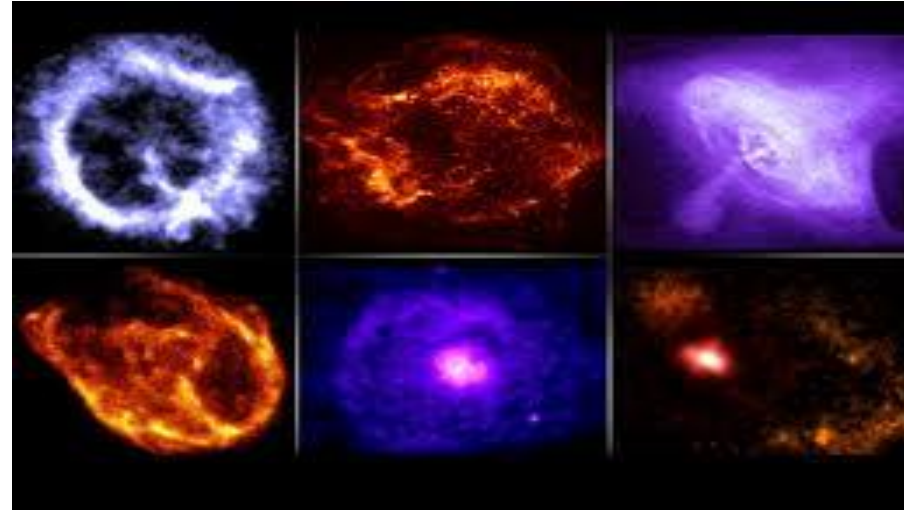


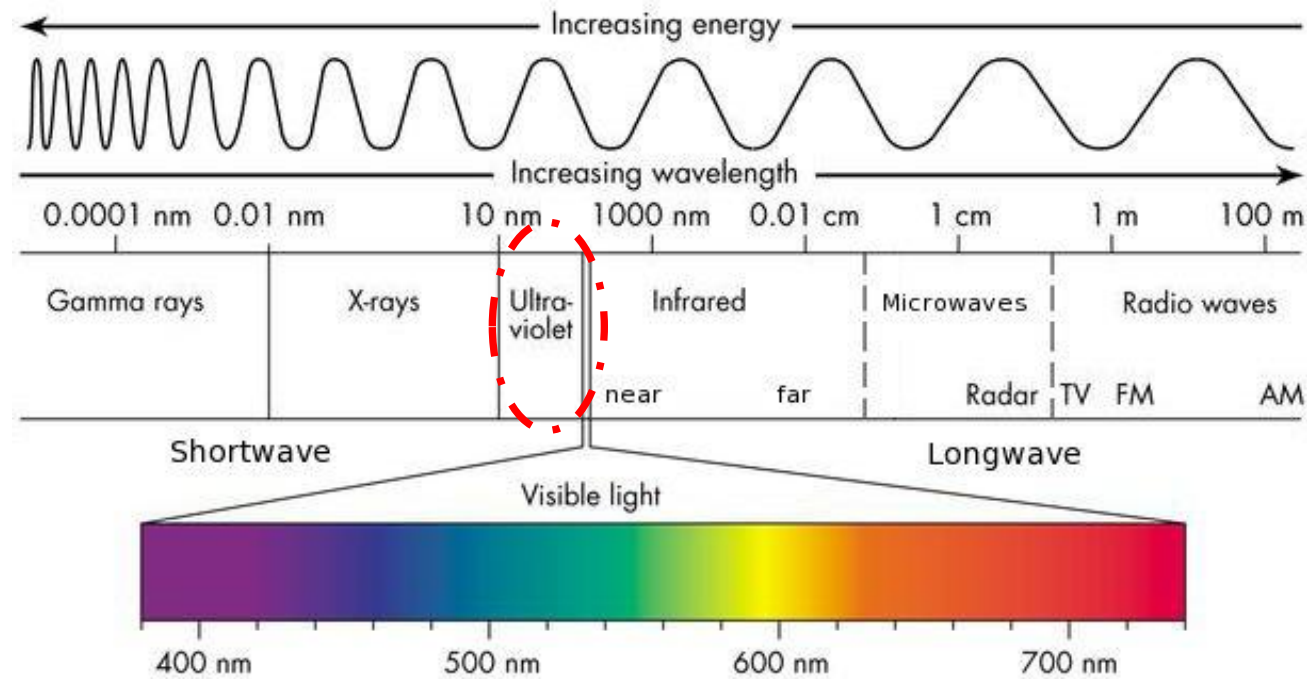
FIGURE 1.7 Examples of X-ray imaging. (a) Chest X-ray. (b) Aortic angiogram. (c) Head CT. (d) Circuit boards. (e) Cygnus Loop. (Images courtesy of (a) and (c) Dr. David R. Pickens, Dept. of Radiology & Radiological Sciences, Vanderbilt University Medical Center, (b) Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, (d) Mr. Joseph E. Pascente, Lixi, Inc., and (e) NASA.)



**X-ray images from the space
The Chandra X-Ray Observatory**



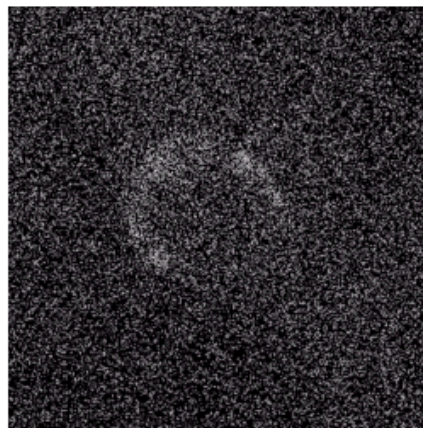
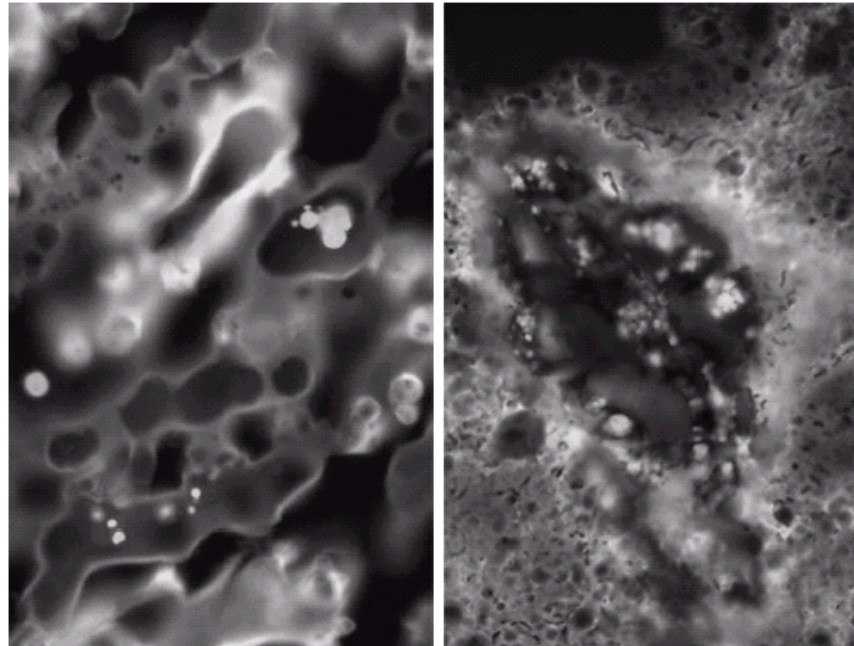
Ultra-violet



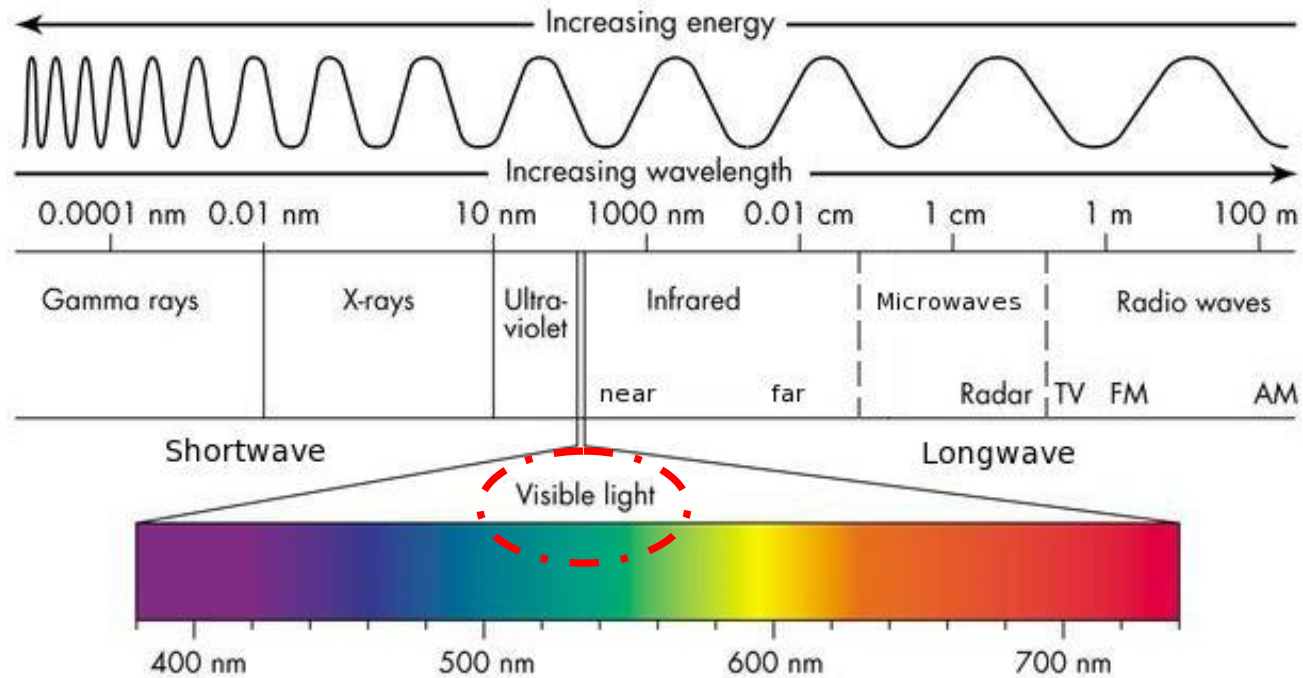
Ultra-violet

a b
c

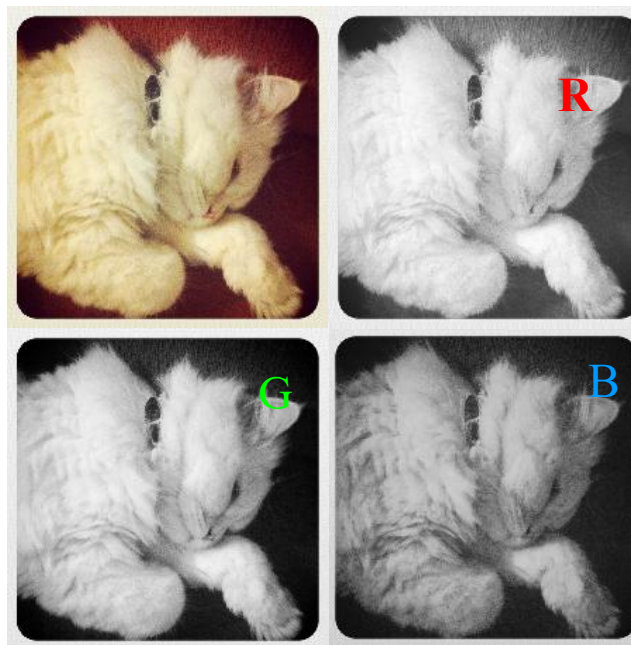
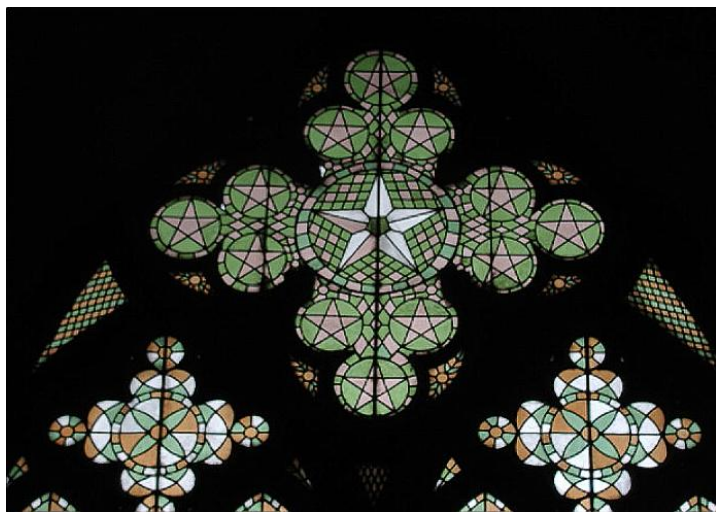
FIGURE 1.8
Examples of
ultraviolet
imaging.
(a) Normal corn.
(b) Smut corn.
(c) Cygnus Loop.
(Images courtesy
of (a) and
(b) Dr. Michael
W. Davidson,
Florida State
University,
(c) NASA.)



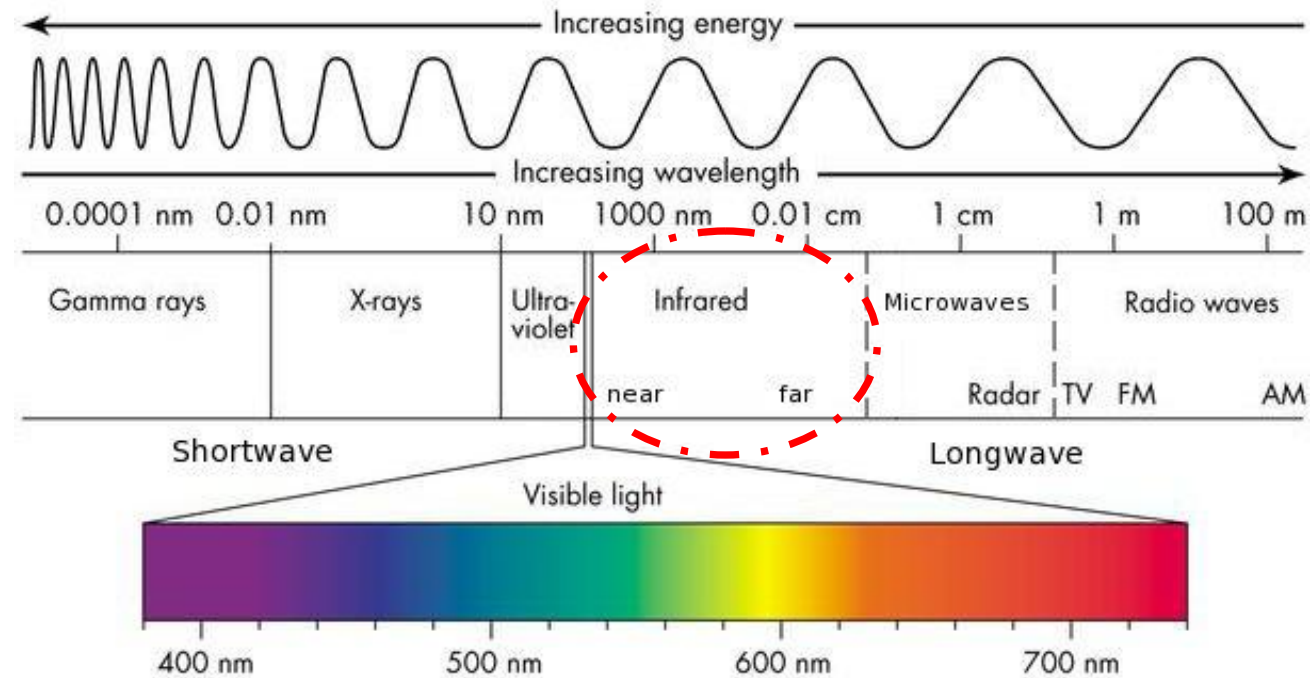
Visible light



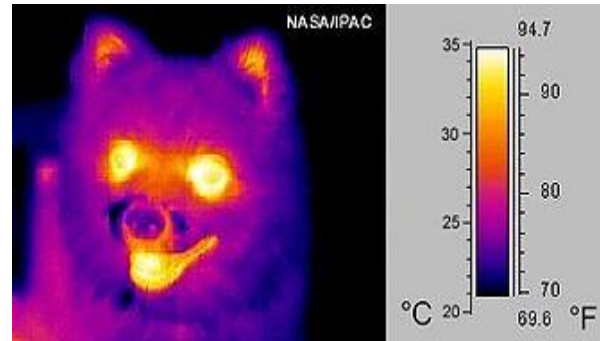
Visible light



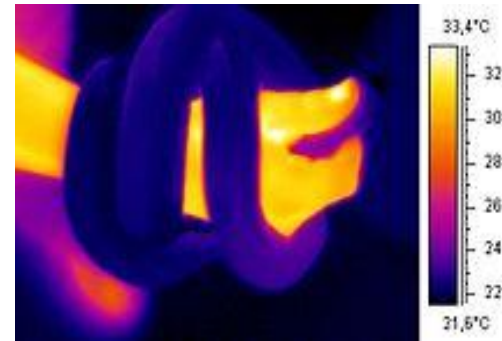
Infrared



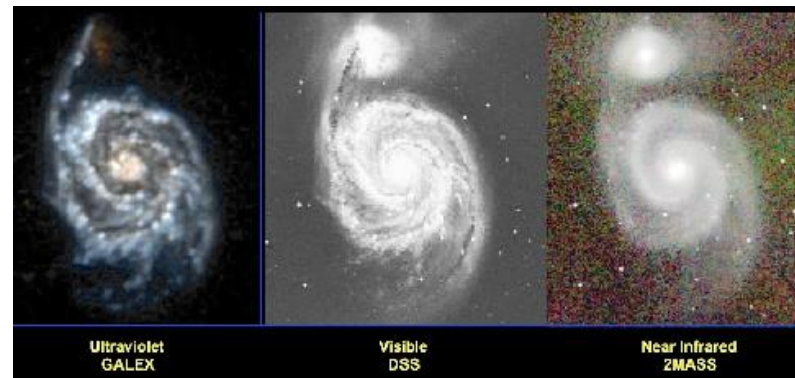
Infrared



infrared ("thermal") image



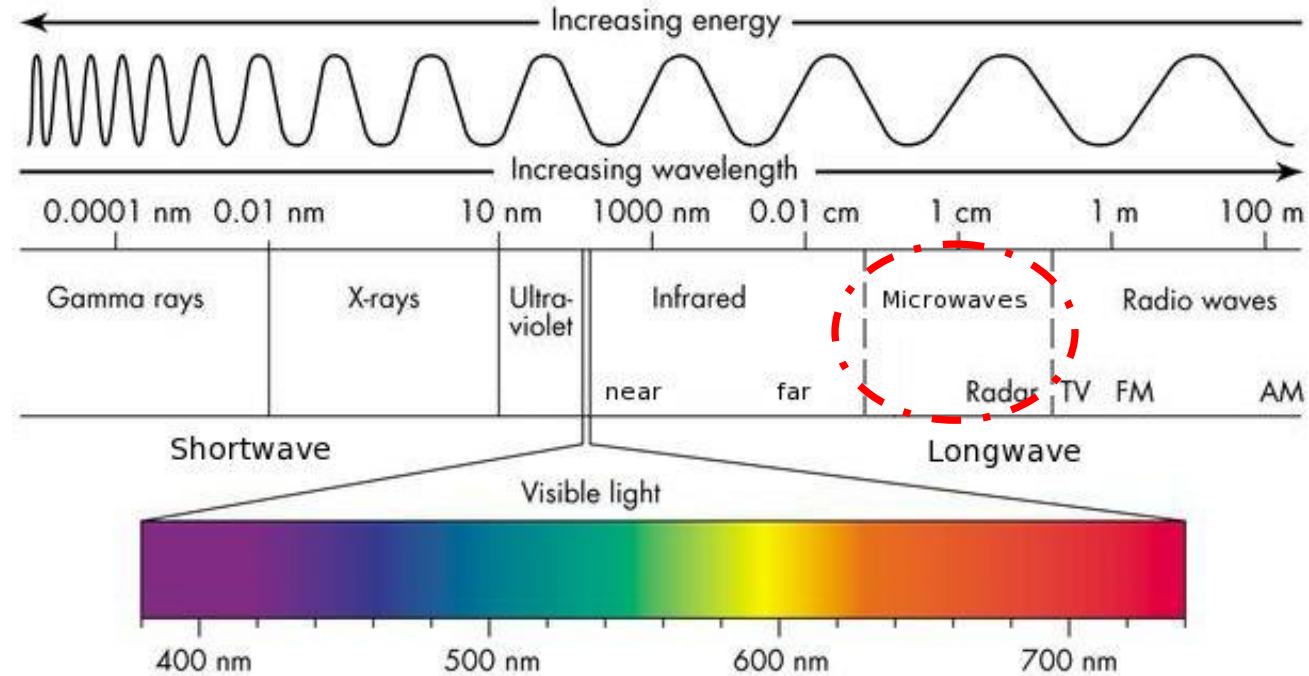
Snake around the arm



Messier 51 in ultraviolet (GALEX), visible (DSS), and **near infrared** (2MASS). *Courtesy of James Fanson.*

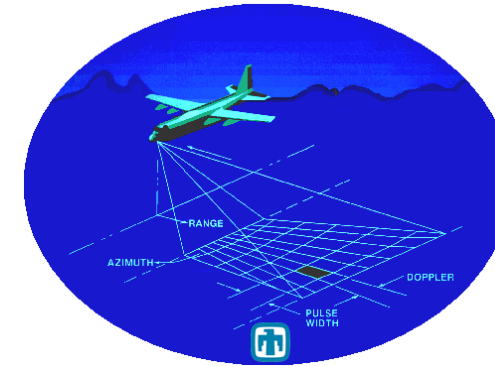
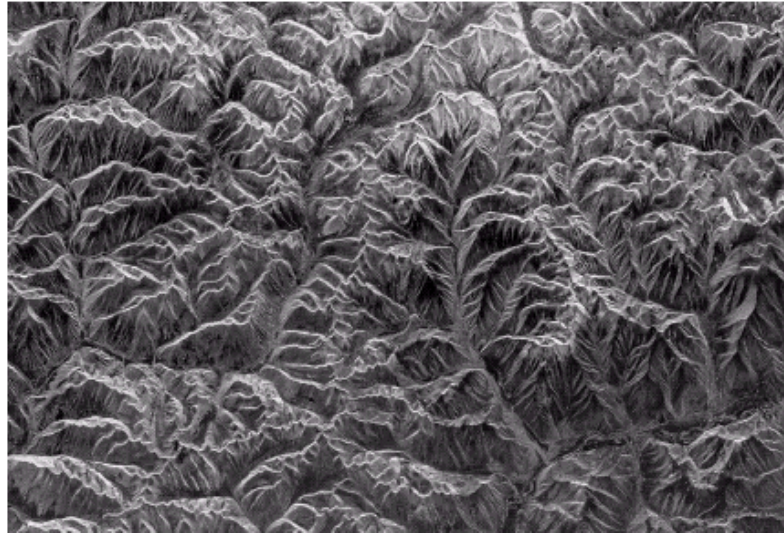


Microwaves



Microwaves

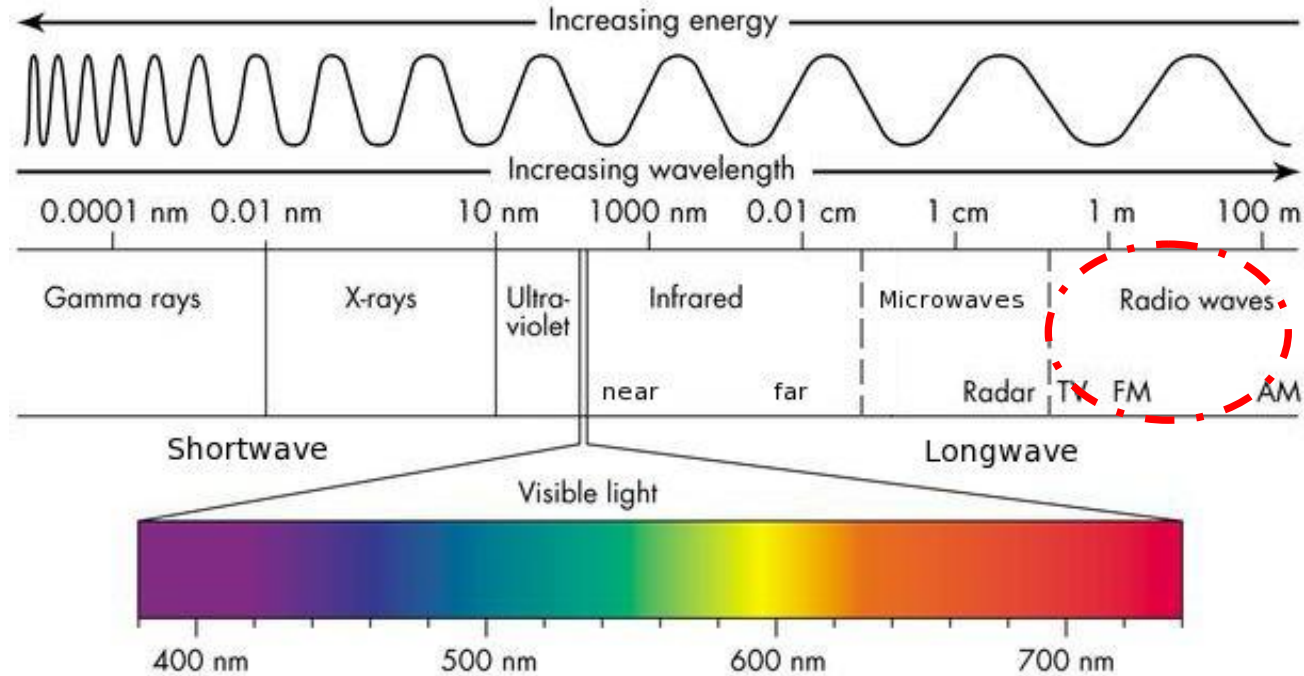
FIGURE 1.16
Spaceborne radar
image of
mountains in
southeast Tibet.
(Courtesy of
NASA.)



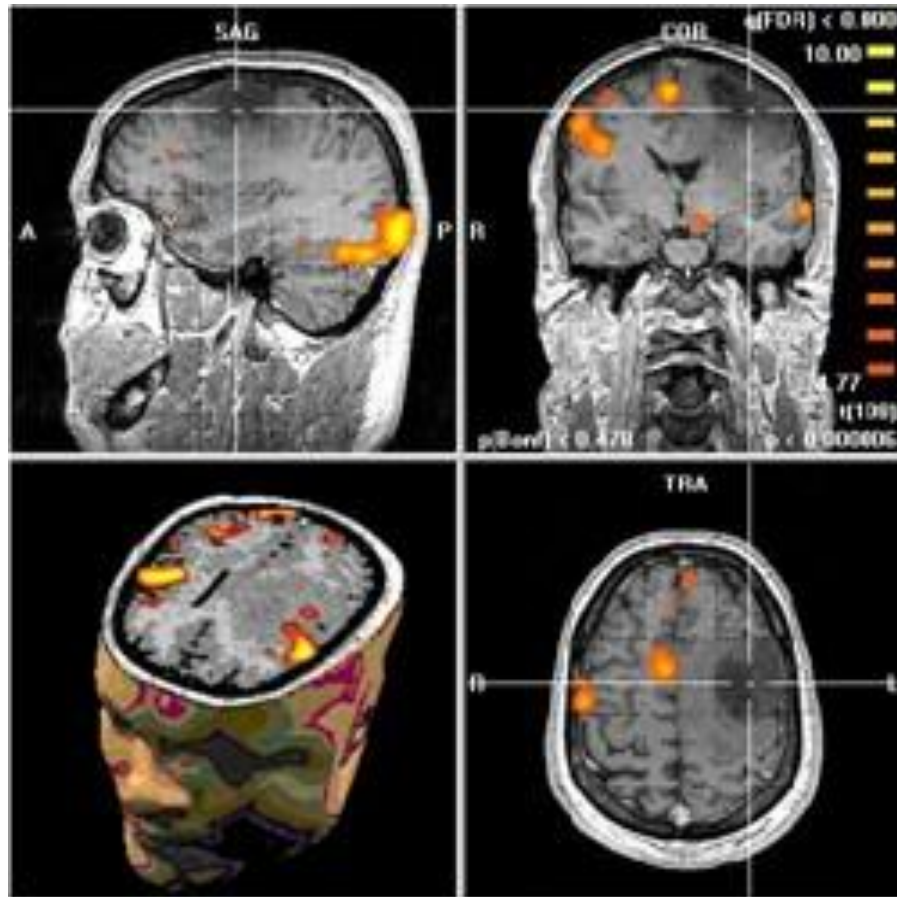
**Synthetic Aperture Radar
System**



Radio Waves



Radio Waves



MRI image slices from the brain



Digital Images based on the EM Spectrum

An example showing Imaging in all of the bands

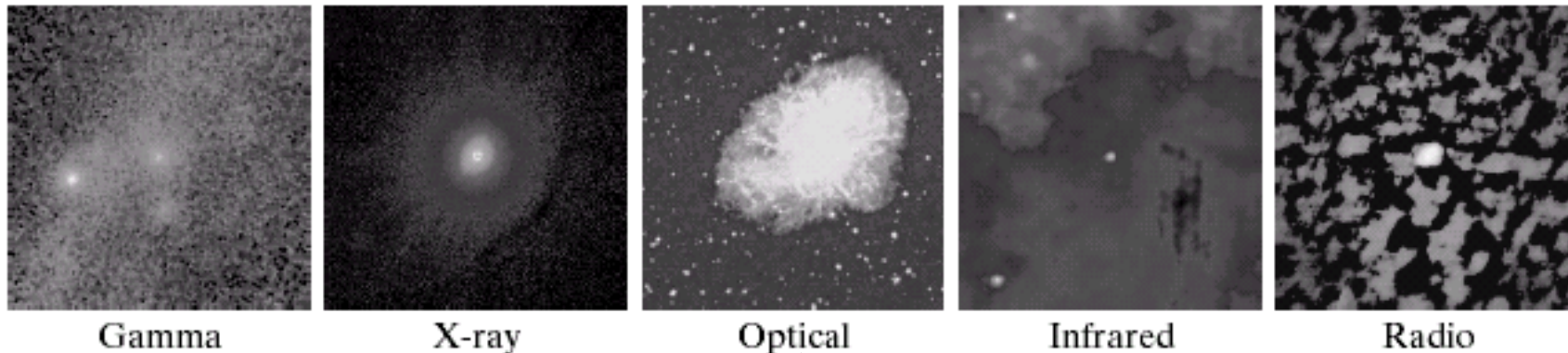
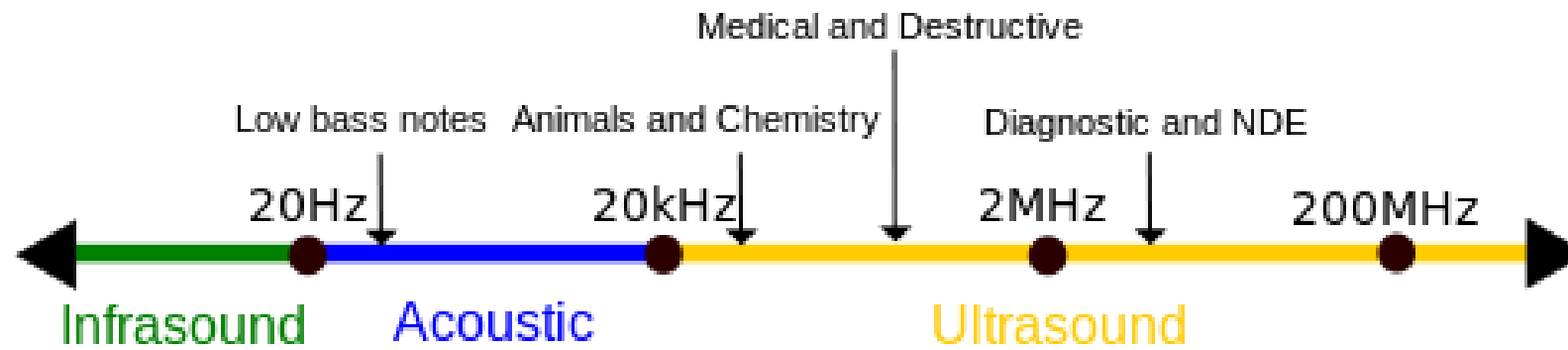


FIGURE 1.18 Images of the Crab Pulsar (in the center of images) covering the electromagnetic spectrum. (Courtesy of NASA.)

Visible light



Ultrasound Imaging



Ultrasonic spectrum



Ultrasoundic Baby image during pregnancy



Ultrasound image acquisition device



The Fundamental Pipeline of Image Processing

When we solve a problem using DIP, we generally follow a structured sequence of steps. Understanding this pipeline is the key to the entire course structure:

1.Acquisition: The first step is always capturing the energy (via a sensor) and digitizing it. Without a high-quality digital signal, processing is limited.

2.Enhancement vs. Restoration:

- Enhancement** is subjective. It makes an image look "better" to a human observer (e.g., brightening a dark photo).
- Restoration** is objective. It attempts to recover an image that has been degraded by using mathematical models of the degradation (e.g., removing blur caused by camera motion).



Continued..

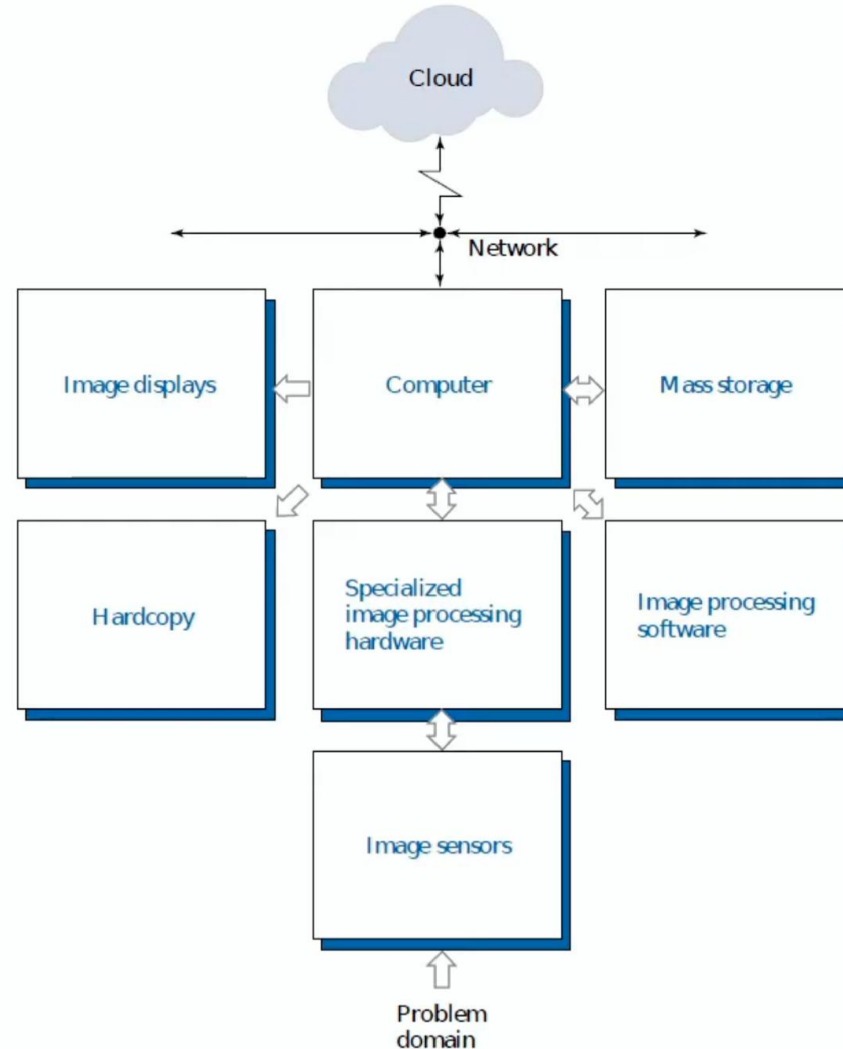
3.Morphological Processing: This deals with the shape of objects. It involves tools to extract image components like boundaries and skeletons, which are useful for describing shape.

4.Segmentation: Often considered the most difficult step. This partitions an image into its constituent parts (e.g., separating a "car" from the "background"). The success of automated recognition relies heavily on accurate segmentation.

5.Recognition: The final stage where the computer assigns a label (like "vehicle" or "tumor") to the object it has segmented and described.

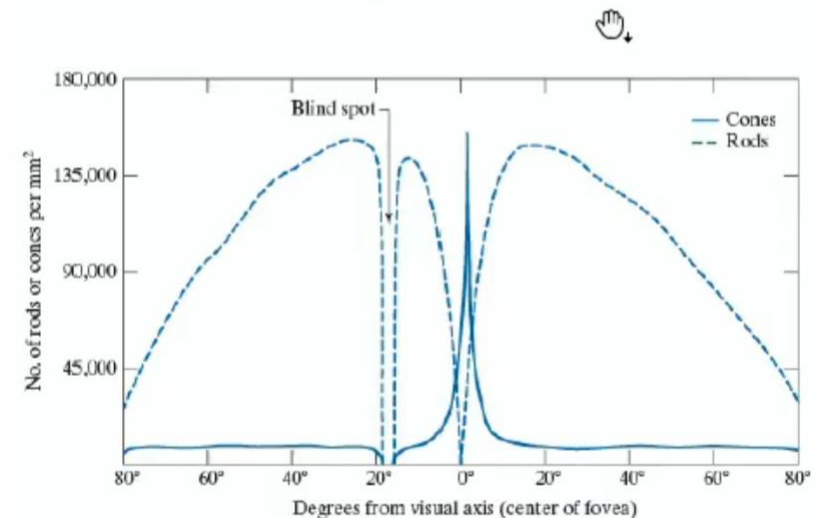
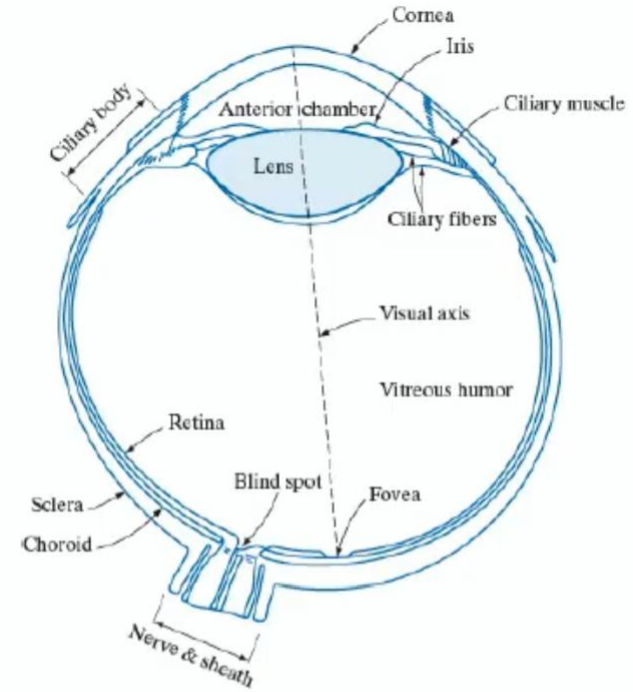


Components of an Image Processing System



Elements of Visual Perception

- **Structure of the Human Eye:**
- Cornea, Iris, Lens, Retina.
- **Retina:** Contains light receptors.
 - **Cones (6-7 million):** Located centrally (fovea), sensitive to color, high detail, photopic (bright light) vision.
 - **Rods (75-150 million):** Distributed over the retina, sensitive to low illumination, not color sensitive, scotopic (dim light) vision.



Elements of Visual Perception

- In regular cameras, the lens has a fixed focal length, and focusing is achieved by varying the distance between the lens and the imaging plane.
- In the human eye, however, the distance between the center of the lens and the retina is fixed (approximately 17 mm), and the focal length is changed by varying the shape of the lens (14–17 mm).

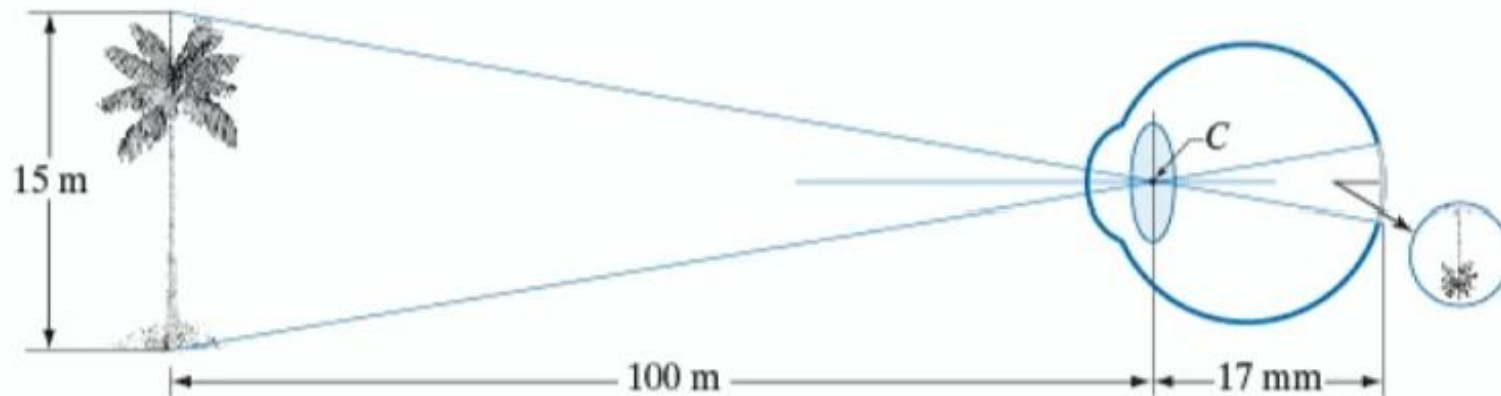


Image Sensing and Acquisition

- Images are generated by the combination of an “illumination” source and the reflection or absorption of energy from the source by the elements of the “scene”.
- Illumination can be from a source of electromagnetic energy, or from less traditional sources such as ultrasound, acoustics, or even computer-generated.
- Scene elements can be familiar objects, or molecules, rock formations, human brain, etc.

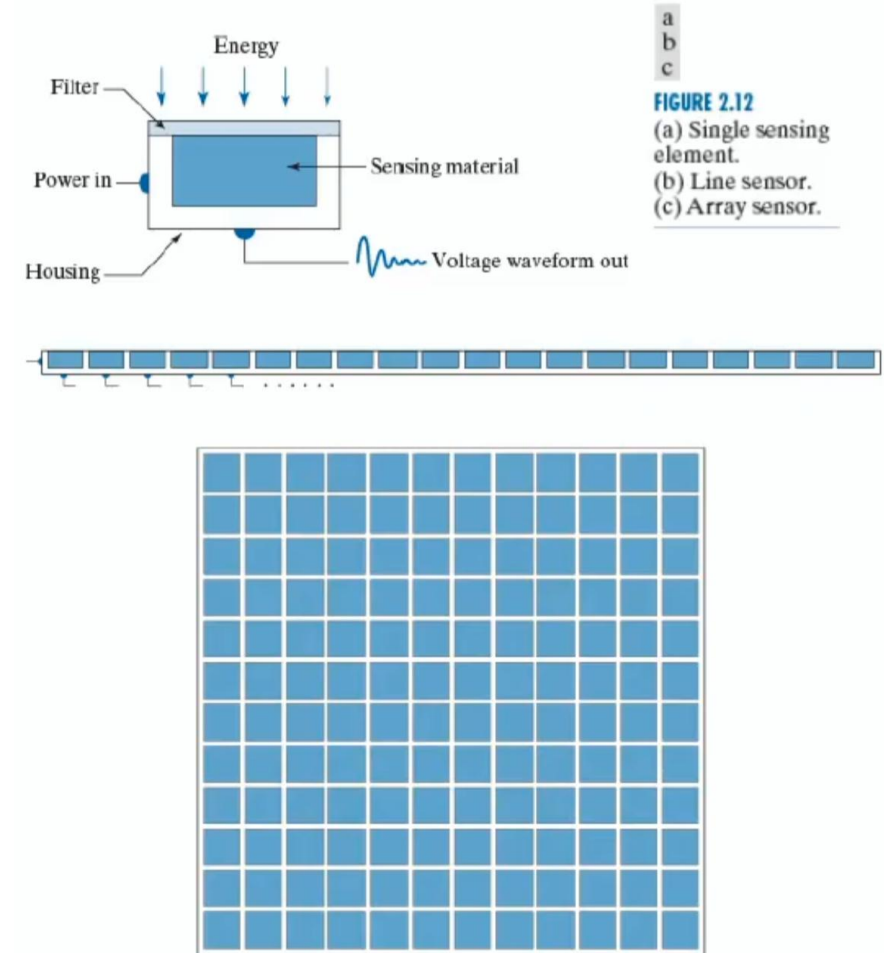
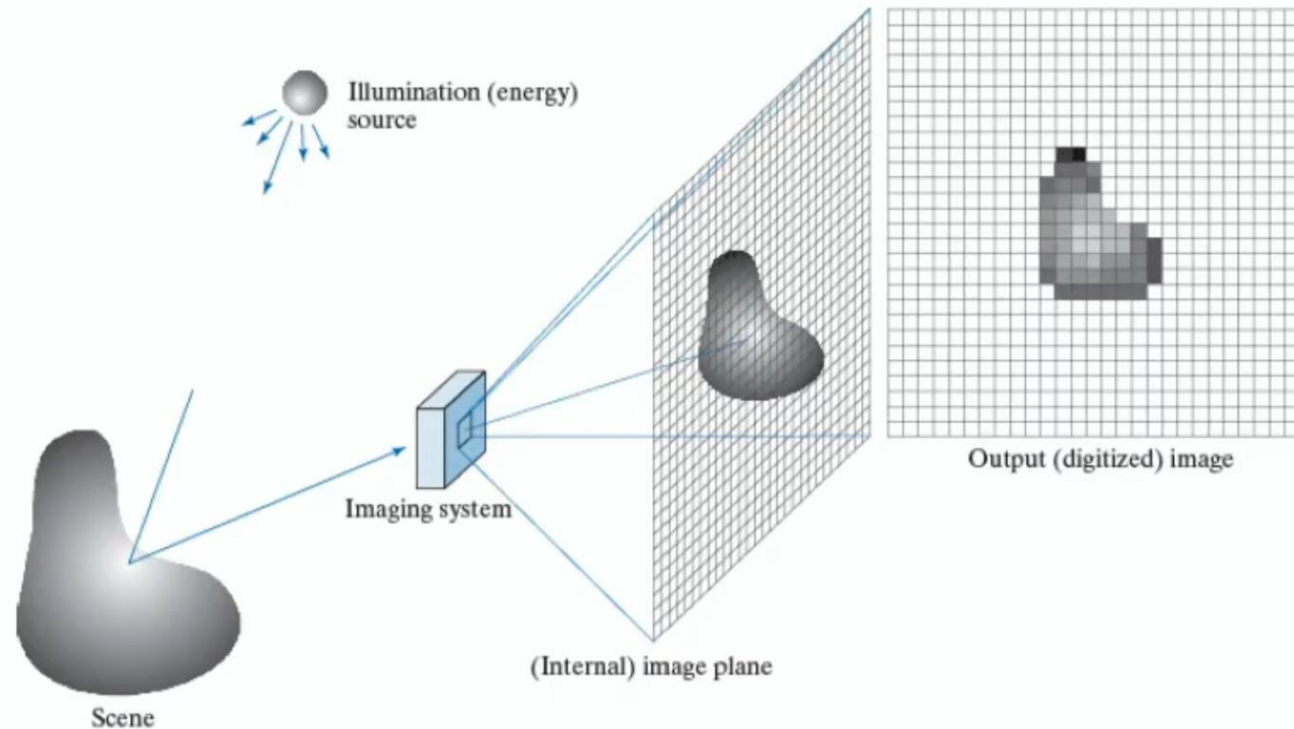


Image Sensing and Acquisition



a
b c d e

FIGURE 2.15 An example of digital image acquisition. (a) Illumination (energy) source. (b) A scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.



Image Formation Model

An image is denoted by a function $f(x, y)$, where the value of f at spatial coordinates (x, y) is a scalar quantity proportional to energy radiated by a physical source.

The values of f are non-negative and finite:
 $0 \leq f(x, y) < \infty$.

Function $f(x, y)$ is characterized by two components:

Illumination: the amount of source illumination incident on the scene being viewed, represented by $i(x, y)$.

Reflectance: the amount of illumination reflected by the objects in the scene, $r(x, y)$.

$$\begin{aligned} f(x, y) &= i(x, y) \times r(x, y) \\ 0 &\leq i(x, y) < \infty \\ 0 &\leq r(x, y) \leq 1 \end{aligned}$$

In some cases, for example X-ray imaging, we have transmissivity instead of reflectance.



Image Sampling and Quantization

Assuming $f(s, t)$ as a continuous image function, using sampling and digitization, we create the image $f(x, y)$, containing M rows and N columns.

The spatial coordinate values are shown by integers as:

$$x = 0, 1, 2, \dots, M - 1 \text{ and}$$

$$y = 0, 1, 2, \dots, N - 1.$$

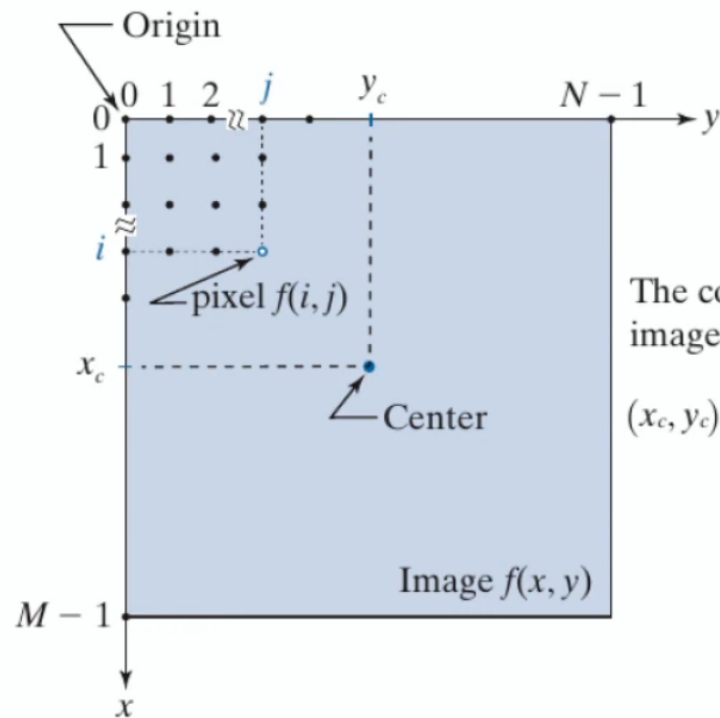
For image $f(x, y)$, we have L number of intensity levels, represented as a power of 2. For example, in an 8-bit image, we have 256 intensity levels:

$$L = 2^k$$



Image Sampling and Quantization

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$



The coordinates of the image center are

$$(x_c, y_c) = \left(\text{floor}\left(\frac{M}{2}\right), \text{floor}\left(\frac{N}{2}\right) \right)$$

$$\mathbf{A} = \begin{bmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,N-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,N-1} \\ \vdots & \vdots & & \vdots \\ a_{M-1,0} & a_{M-1,1} & \cdots & a_{M-1,N-1} \end{bmatrix}$$



Spatial and Intensity Resolution

- **Spatial resolution:** the size of the smallest perceptible details in an image.
- May be measured by the number of pixels per unit distance.
- Spatial resolution is dependent on the sampling rate.



a b
c d

FIGURE 2.23
Effects of
reducing spatial
resolution. The
images shown
are at:
(a) 930 dpi,
(b) 300 dpi,
(c) 150 dpi, and
(d) 72 dpi.

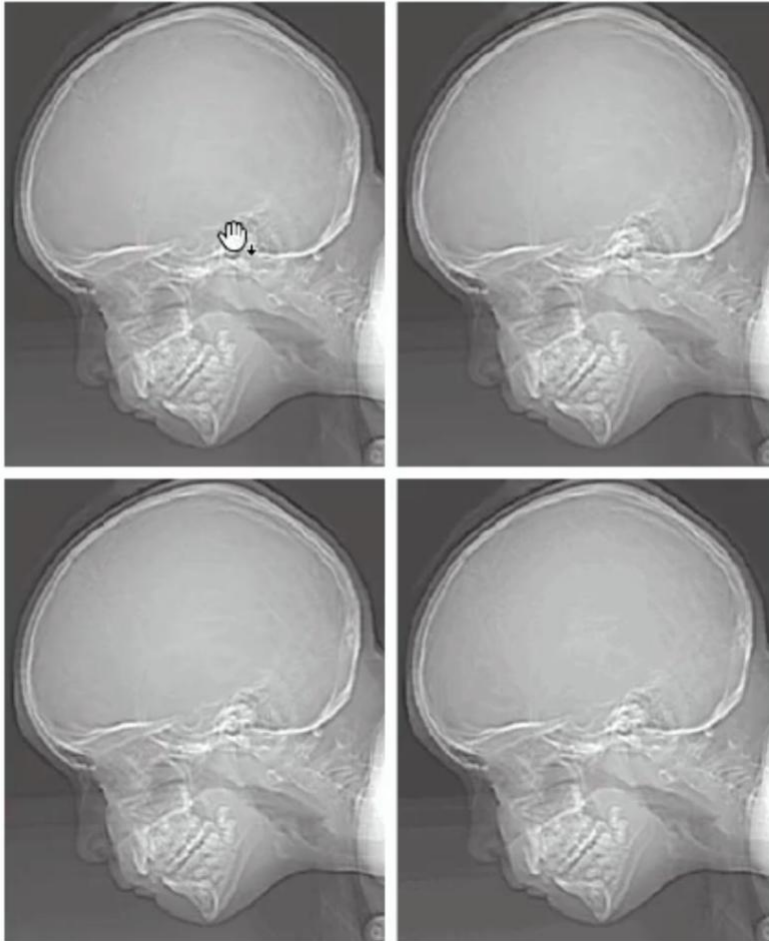


Spatial and Intensity Resolution

a b
c d

FIGURE 2.24

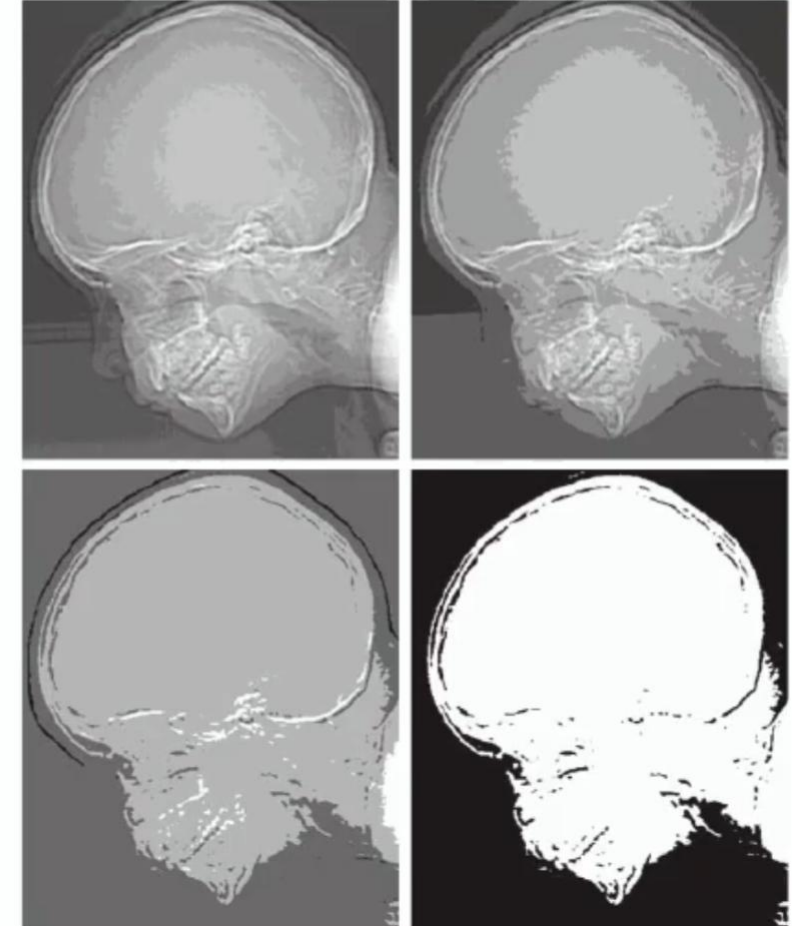
(a) 774×640 , 256-level image. (b)-(d) Image displayed in 128, 64, and 32 intensity levels, while keeping the spatial resolution constant. (Original image courtesy of the Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)



e f
g h

FIGURE 2.24

(Continued)
(c)-(h) Image displayed in 16, 8, 4, and 2 intensity levels.



Intensity resolution: the smallest discernible change in the intensity level. Measured in the number of bits used for quantization.



Spatial and Intensity Resolution

Both are digitization-dependent:

- Spatial resolution depends on the number of samples (N).
- Intensity resolution depends on the number of bits (k).

Different artifacts:

- Too low spatial resolution results in jagged lines.
- Too low intensity resolution results in false contouring.

Sensitivity:

- Spatial resolution is more sensitive to shape variations.
- Intensity resolution is more sensitive to lighting variations.



Spatial and Intensity Resolution

- **Left:** Least geometric details but more lighting information
- **Middle:** More details but less lighting information
- **Right:** Most geometric details but least lighting information
- What would be good digitization schemes for these images?



Spatial and Intensity Resolution

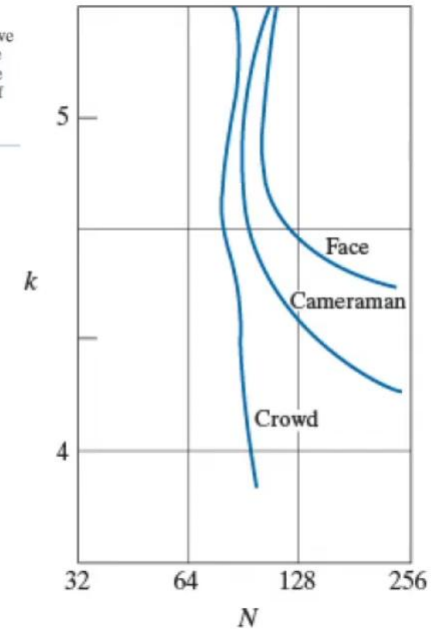
- **The iso-preference curves**

- Change N (sampling) and k (intensity) values and compare the quality of the images obtained.
- Each curve shows the images with the same quality judged by observers.

- **We can see:**

- Images with more shape detail (e.g., crowd) need fewer intensity levels to achieve the same quality.
- Images with less shape detail (e.g., face) are more sensitive to the intensity resolution but less sensitive to the spatial resolution.

FIGURE 2.26
Representative
isopreference
curves for the
three types of
images in
Fig. 2.25.



Thank You

